

Ultrasonic-Based Anti-Theft Alarm System

Arduino UNO · HC-SR04 Ultrasonic Sensor · LED Indicators · Buzzer Alert

Arduino Project

Embedded Systems

Security & IoT

The Ultrasonic-Based Anti-Theft Alarm System is an Arduino-powered security device that uses sound waves to detect nearby objects or intruders. When armed, it continuously measures distance — triggering a buzzer alarm and flashing red LED when anything moves within 200 cm. A green LED confirms the system is active and clear.

Arm / Disarm

Push-button toggle arms or disarms the entire system instantly

Real-Time Detection

HC-SR04 pings 20×/sec with a 200 cm detection range

Dual Alerts

1 kHz buzzer tone + flashing red LED on intrusion detection

Arduino UNO

Microcontroller

ATmega328P, 14 digital I/O pins, runs at 16 MHz. The brain of the system.

 Power: 5V & GND

HC-SR04

Ultrasonic Sensor

Sends 40 kHz pulses; measures echo return time to calculate distance (2–400 cm).

 TRIG→D7, ECHO→D6

Buzzer (Passive)

Audio Alert

Piezoelectric buzzer driven by tone() at 1000 Hz when an intruder is detected.

 Signal→D12

Red LED + 330Ω

Visual Alert

Flashes at 200 ms intervals when armed and an object enters the detection zone.

 Anode→D11

Green LED + 330Ω

Status Indicator

Stays solid ON when the system is armed and no threat is detected.

 Anode→D10

Push Button

Arm/Disarm Toggle

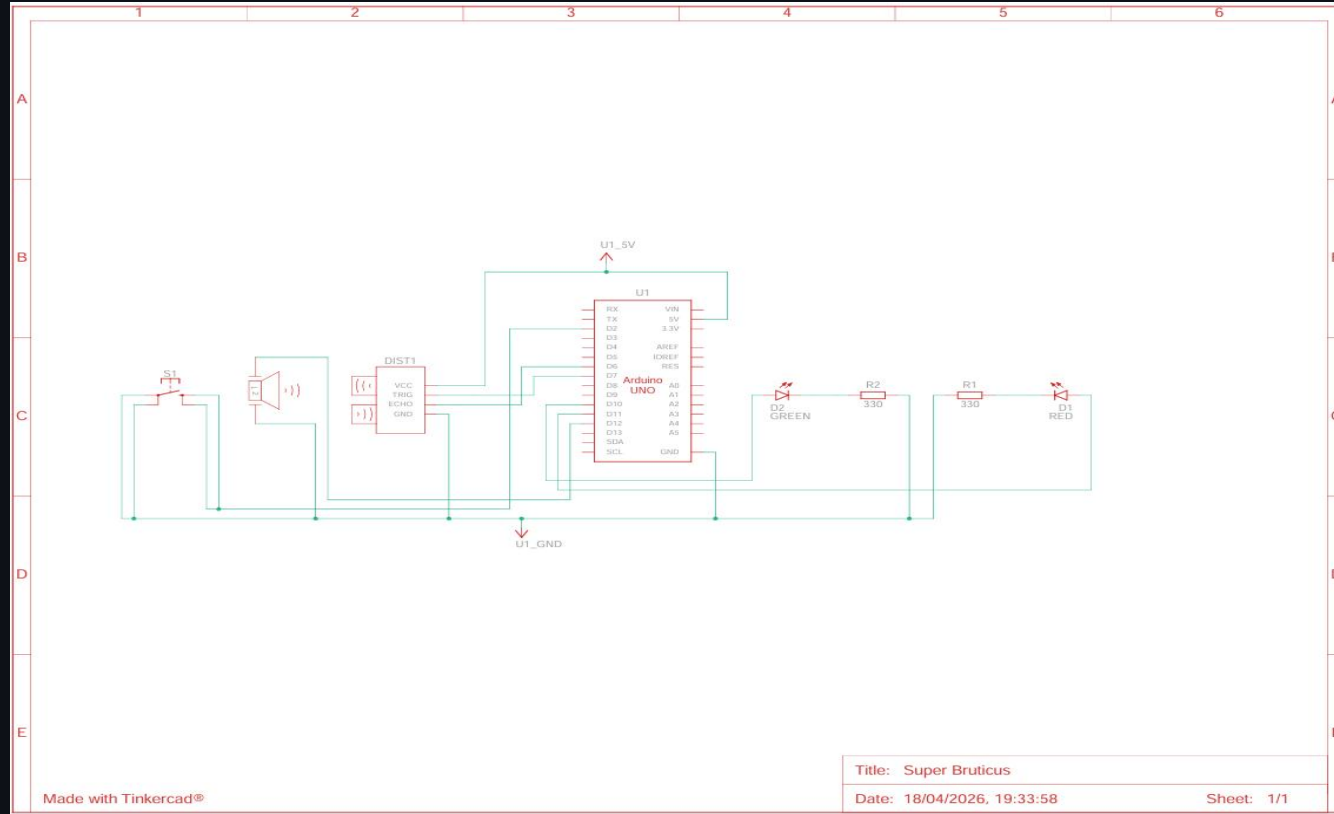
Connected with INPUT_PULLUP. Single press toggles system state; 300 ms debounce.

 Signal→D2

Component	Arduino Pin	Wire Color	330Ω Resistors
HC-SR04 VCC	5V	Red	Each LED must use a 330 Ω resistor in series to limit current to ~10 mA and protect the Arduino pin.
HC-SR04 GND	GND	Black	
HC-SR04 TRIG	D7	Yellow	
HC-SR04 ECHO	D6	Green	INPUT_PULLUP Button uses internal pull-up; LOW = pressed. Eliminates need for external resistor.
Buzzer (+)	D12	Purple	
Red LED (+)	D11 via 330Ω	Red	
Green LED (+)	D10 via 330Ω	Green	Breadboard GND All GND connections share the breadboard rail tied to Arduino GND.
Button Pin 1	D2	Orange	
Button Pin 2	GND	Black	

Schematic Diagram

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Arduino UNO

HC-SR04 Sensor

Buzzer

LEDs + Resistors

1

Button Press

User presses button →
systemArmed toggles.
300 ms debounce
prevents double-trigger.

2

Ultrasonic Ping

TRIG goes HIGH for 10 μ s
→ sensor fires 8 x 40 kHz
pulses toward target.

3

Echo Timing

pulseIn() measures how
long ECHO stays HIGH
(μ s). Speed of sound =
0.0343 cm/ μ s.

4

Distance Calc

distance = duration ×
0.0343 ÷ 2. Divides by 2
because pulse travels TO
object and back.

5

Threat Check

If distance < 100 cm →
tone(buzzer,1000) + red
LED flash. Else → green
LED on, silence.

Code Walkthrough

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● ● ● sketch.ino

```
#define trig 7  echo 6  buzzer 12
#define greenLED 10  redLED 11  button 2

bool systemArmed = false;

void setup() {
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(redLED, OUTPUT);
  pinMode(button, INPUT_PULLUP);
}

void loop() {
  // Toggle system ON/OFF when button pressed
  if (digitalRead(button) == LOW) {
    systemArmed = !systemArmed; // switch state
    delay(300); // debounce delay
  }
```

```
  if (!systemArmed) {
    // SYSTEM OFF
    noTone(buzzer);
    digitalWrite(redLED, LOW);
    digitalWrite(greenLED, LOW);
    return; // skip rest
  }

  // SYSTEM ON → run sensor
  long duration;
  float distance;

  digitalWrite(trig, LOW);
  delayMicroseconds(2);

  digitalWrite(trig, HIGH);
  delayMicroseconds(10);

  digitalWrite(trig, LOW);

  duration = pulseIn(echo, HIGH);
  distance = duration * 0.0343 / 2;

  if (distance < 100) {
    tone(buzzer, 1000);

    digitalWrite(greenLED, LOW);

    digitalWrite(redLED, HIGH);
    delay(200);
    digitalWrite(redLED, LOW);
    delay(200);
  } else {
    noTone(buzzer);

    digitalWrite(redLED, LOW);
    digitalWrite(greenLED, HIGH);
  }

  delay(50);
}
```

What We Built

- Arduino UNO as the central controller
- HC-SR04 sensor measuring distance in real-time
- Button to toggle armed / disarmed state
- Red LED + buzzer for intrusion alert
- Green LED for armed + clear confirmation
- Resistor-protected LEDs for circuit safety
- Clean loop() with no blocking beyond debounce

Circuit Diagram

